Class XII Session 2025-26
Subject - Chemistry
Sample Question Paper - 2

Time Allowed: 30 Minutes

General Instructions:

Read the following instructions carefully.

1. There are 16 questions in this question paper with internal choice.
2. SECTION A consists of 16 multiple-choice questions carrying 1 mark each.
3. **All questions are compulsory.**
4. **Use of log tables and calculators is not allowed.**

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Section A

1. Ethylidene chloride is a/an _____. **[1]**

   a) gem-dihalide
   b) vic-dihalide
   c) vinylic halide
   d) allylic halide

2. Which of the following bases is not present in DNA? **[1]**

   a) Adenine
   b) Cytosine
   c) Thymine
   d) Uracil

3. Many naturally occurring aldehydes and ketones are used in the blending of perfumes and flavouring agents. But the preferred ones are_____. **[1]**

   a) lower ketones
   b) higher aldehydes
   c) lower aldehydes
   d) higher ketones

4. The correct IUPAC name of $CH_3 - C(CH_3)(OH) - CH_2CH_3$ is **[1]**

   a) 2,2-Dimethylpropanol
   b) 2-Methylbutan-2-ol
   c) tert-butyl alcohol
   d) 3-Methylbutan-3-ol

5. When a catalyst increases the rate of a chemical reaction, then the rate constant (k): **[1]**

   a) increases
   b) may increase or decrease depending on the order of the reaction

c) remains constant

d) decreases

6. Match the items given in column I with that in column II: **[1]**

Column I	Column II
(a) Ω^{-1} cm^2mol^{-1}	(i) Mercury Cell
(b) $\alpha_{dissoiation}$ is low	(ii) λ_m
(c) Decreases with dilution	(iii) Weak Electrolyte
(d) Electrolyte is a paste of KOH/ZnO	(iv) Conductivity

a) (a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)

b) (a) - (i), (b) - (ii), (c) - (iii), (d) - (iv)

c) (a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)

d) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

7. Maximum number of molecules of CH_3I that can react with a molecule of CH_3NH_2 is: **[1]**

a) 4

b) 3

c) 1

d) 2

8. The action of nitrous acid on ethylamine gives mainly: **[1]**

a) ethyl nitrite

b) ethane

c) ethyl alcohol

d) nitroethane

9. Rate of ionic reactions are generally **[1]**

a) Very slow

b) Moderate

c) Slow

d) Very fast

10. Which of the following does not give aldol condensation reaction? **[1]**

a)

b)

c) CH_3COCH_3

d) CH_3 - CHO

11. Monochlorination of toluene in sunlight followed by hydrolysis by aq. NaOH yields **[1]**

a) m-cresol

b) benzyl alcohol

c) o-cresol

d) 2,4-dihydroxytoluene

12. Propanamide on reaction with bromine in aqueous NaOH gives: **[1]**

a) Ethanamine

b) Propanenitrile

c) Propanamine

d) N-Methyl ethanamine

13. **Assertion (A):** Except glycine, all naturally occurring α-amino acids are optically active. **[1]**

Reason (R): All α-amino acids occurring naturally except glycine has at least one asymmetric carbon.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

14. **Assertion (A):** Benzoic acid does not give Friedel-Crafts reaction. **[1**

Reason (R): The carboxyl group is deactivating and gets bonded to Lewis acid $AlCl_3$.

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a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

15. **Assertion (A):** Electron withdrawing groups in aryl halides decrease the reactivity towards nucleophilic substitution. **[1]**

Reason (R): 2,4 -Dinitrochlorobenzene is more reactive than chlorobenzene.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

16. **Assertion (A):** The reaction of the alcohol with $SOCl_2$ is catalyzed by the presence of a tertiary amine (R_3N). **[1]**

Reason (R): Tertiary amine promotes the reaction by reacting with the by-product HCl.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

C) A is true but R is false.

d) A is false but R is true.

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